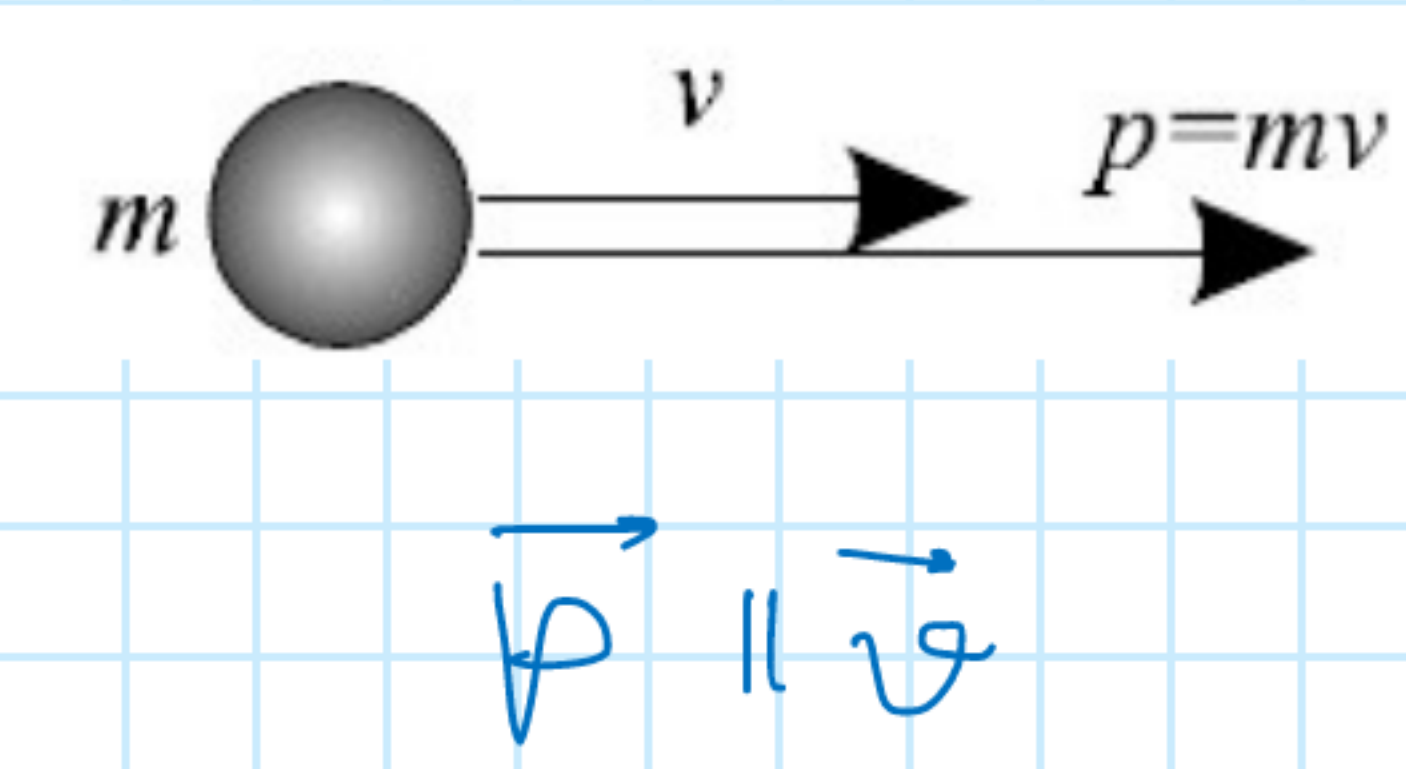


Ligjerata 4ç
Sasia e lëvizjes

$$p = m \cdot v = \hbar \gamma \frac{u}{c}$$
$$v \ll c$$



$$v \leq c$$
$$m \neq \text{const}$$
$$p = m \cdot v = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Ligji i ruajtjes sasisë së lëvizjes

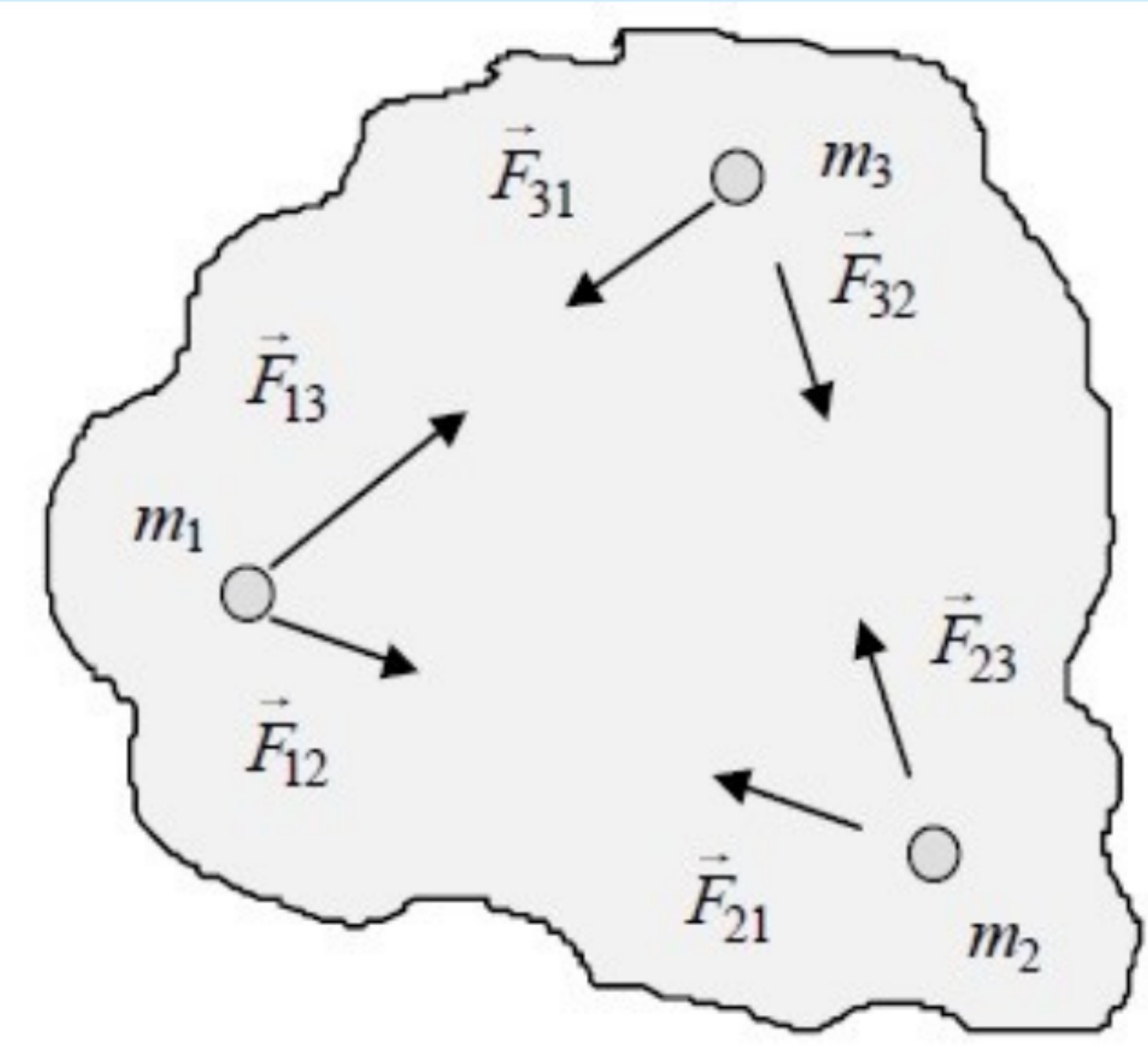
$$F = \frac{dp}{dt}$$

$$\frac{d\vec{p}_1}{dt} = \vec{F}_{21} + \vec{F}_{31} + \vec{F}_{41} + \dots + \vec{F}_{n1}$$

$$\frac{d\vec{p}_2}{dt} = \vec{F}_{12} + \vec{F}_{32} + \vec{F}_{42} + \dots + \vec{F}_{n2}$$

$$\frac{d\vec{p}_3}{dt} = \vec{F}_{13} + \vec{F}_{23} + \vec{F}_{43} + \dots + \vec{F}_{n3}$$

$$\frac{d\vec{p}_n}{dt} = \vec{F}_{1n} + \vec{F}_{2n} + \vec{F}_{3n} + \dots + \vec{F}_{(n-1)n}$$



$$\frac{d}{dt} (\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n) = \underbrace{(\vec{F}_{12} + \vec{F}_{21})}_{=0} + \underbrace{(\vec{F}_{13} + \vec{F}_{31})}_{=0} + \underbrace{(\vec{F}_{14} + \vec{F}_{41})}_{=0} + \dots + \underbrace{(\vec{F}_{1n} + \vec{F}_{n1})}_{=0}$$

$$\frac{d}{dt} (\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n) = 0$$

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n = \text{const}$$

M → masa e revolverit
m → masa e projektilit.



$$\sum_{i=1}^n \vec{p}_i = \sum_{i=1}^n m_i \vec{v}_i = M \vec{v}_0 + m \vec{v}_0 = \text{const.}$$

$$v_0 = 0$$

$$\sum_{i=1}^n \vec{p}_i = \sum_{i=1}^n m_i \vec{v}_i = M \vec{v}_1 + m \vec{v} = \text{const} = 0$$

$$M \vec{v}_1 = -m \vec{v}$$
$$v_1 = \frac{m}{M} \cdot v$$